



Habib University
shaping futures

WATER : SCIENCE, SOCIETY & POLICY

ENVS/SDP 251 L1

Spring Semester 2024

Instructor: Usman Salahuddin

Hours: Tues – Thurs

(3:05 PM – 4:05 PM)

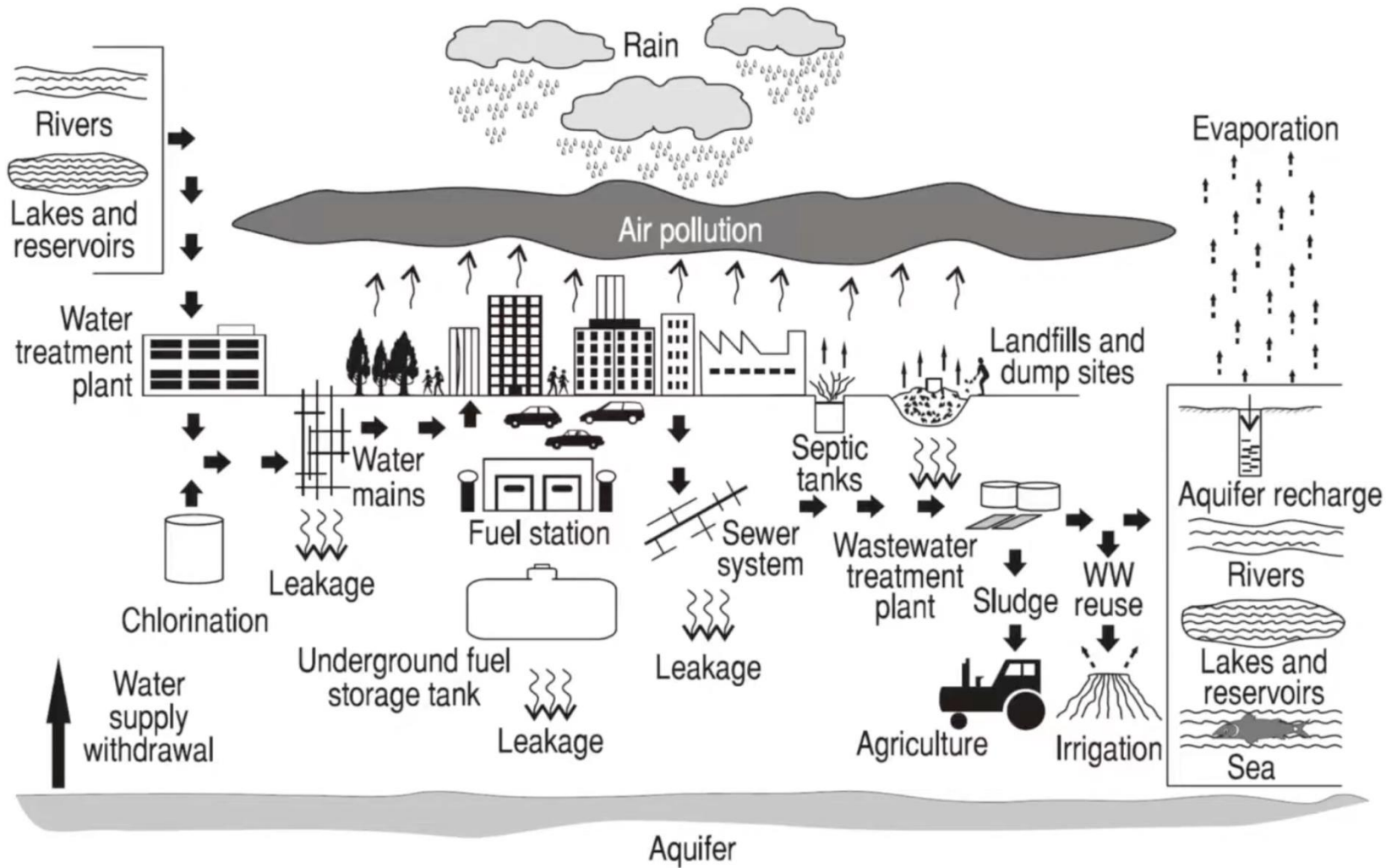
Class Location W-242

Office hour: Wed 1-3PM

Office: C-103

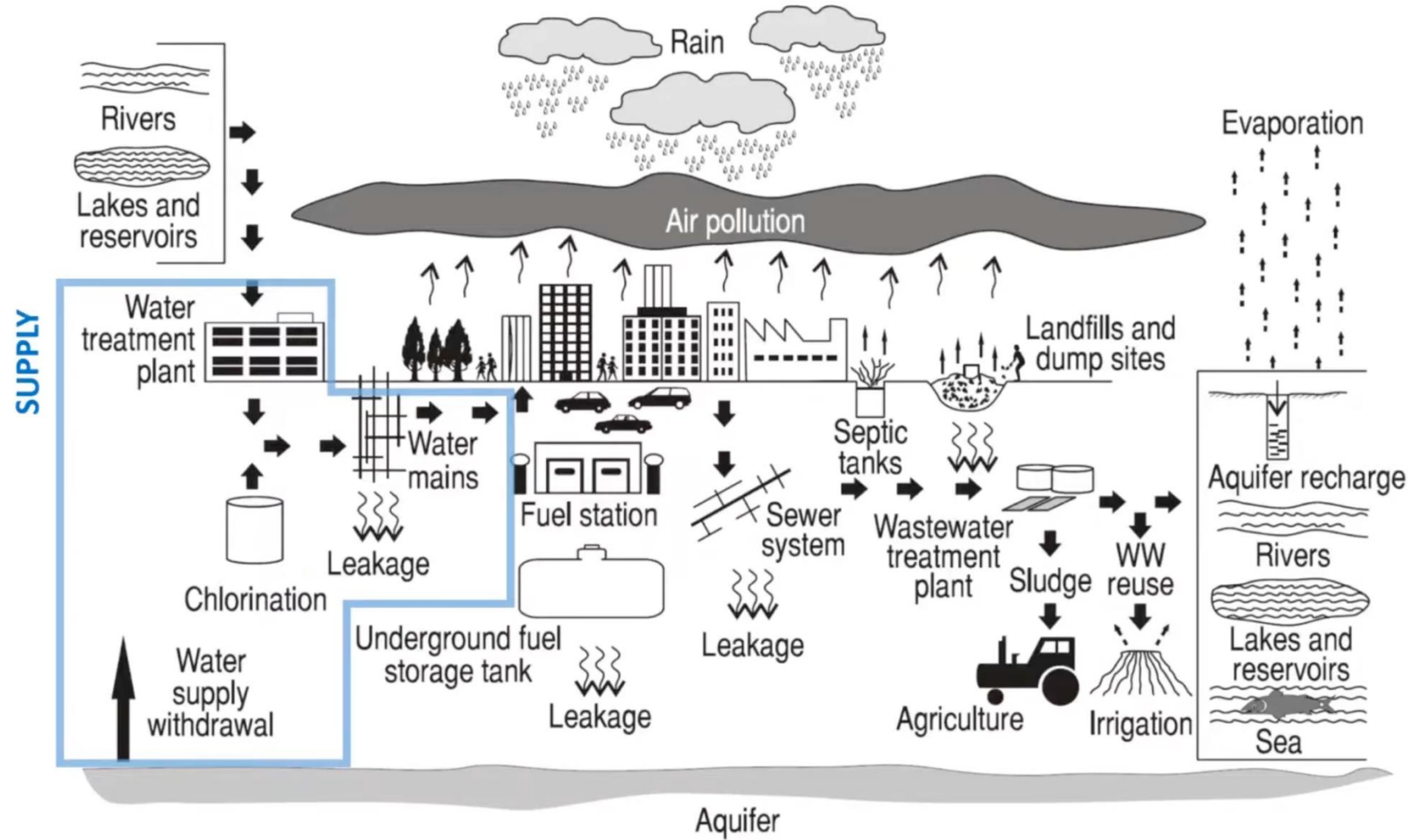
Email: usman.salahuddin@sse.habib.edu.pk

URBAN WATER MANAGEMENT



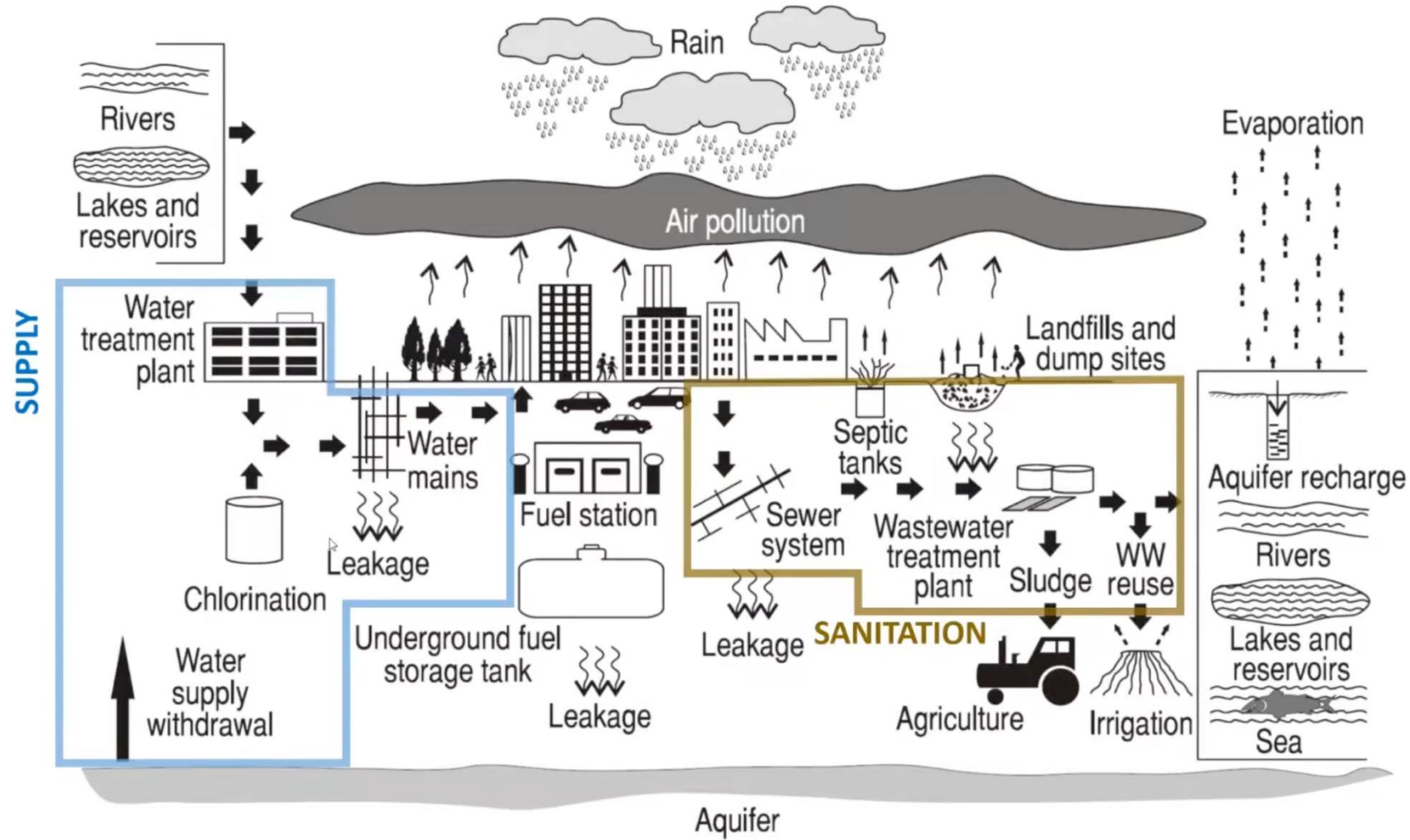
Source: Marsalek, J., Cisneros, B.J., Karamouz, M., Malmquist, P.A., Goldenfum, J.A. and Chocat, B., 2008. Urban water cycle processes and interactions: Urban Water Series-UNESCO-IHP (Vol. 2). CRC Press.

URBAN WATER MANAGEMENT

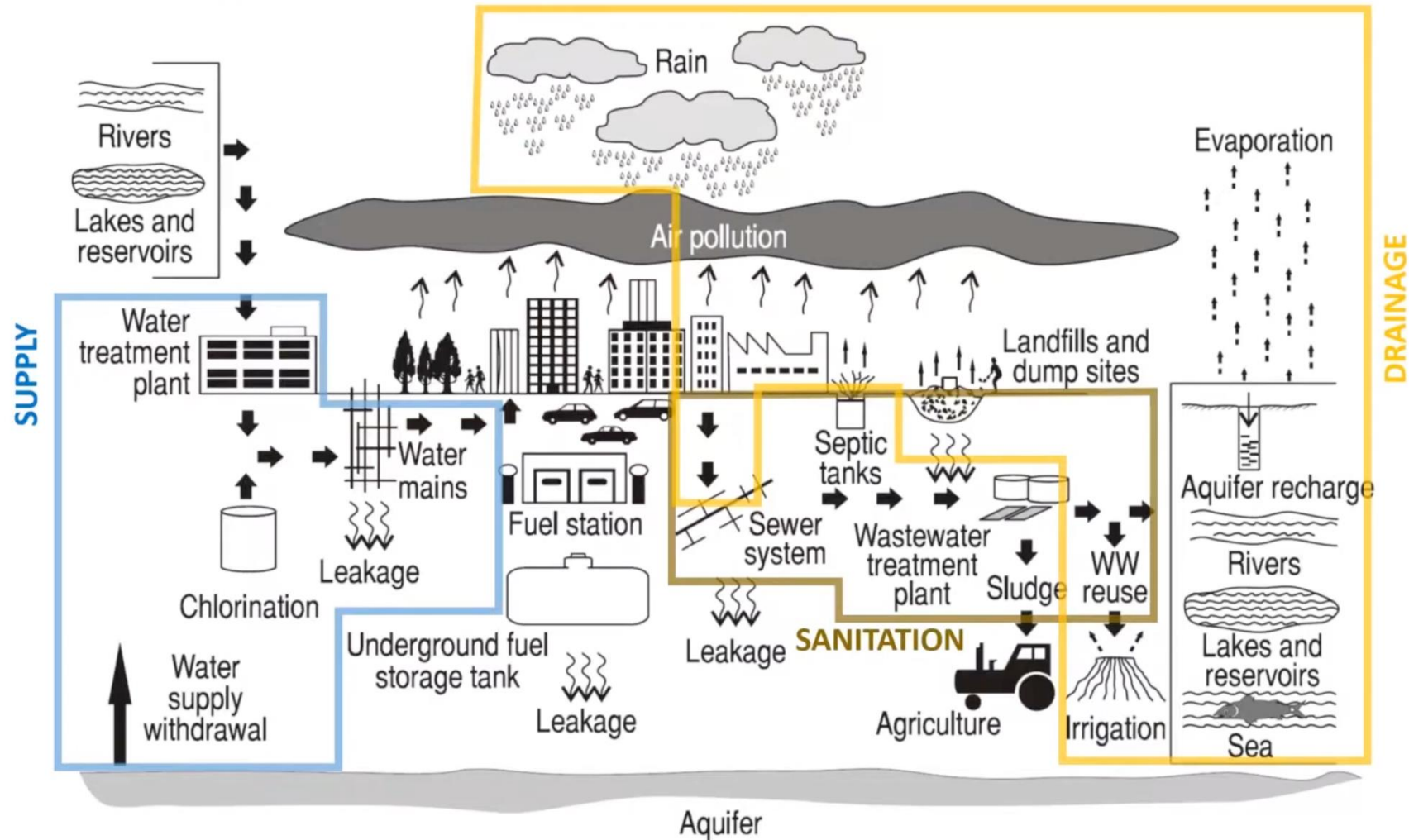


Source: Marsalek, J., Cisneros, B.J., Karamouz, M., Malmquist, P.A., Goldenfum, J.A. and Chocat, B., 2008. Urban water cycle processes and interactions: Urban Water Series-UNESCO-IHP (Vol. 2). CRC Press.

URBAN WATER MANAGEMENT



URBAN WATER MANAGEMENT



URBAN WATER SYSTEM



- **SUPPLY:** Securing a safe and accessible supply of clean water for drinking and other uses – typically obtained from a nearby river, lake or groundwater catchment, desalination of sea water



- **SANITATION:** Collect the unused or waste water and treat it before discharge to the surrounding environment to protect receiving water bodies from pollutants and to protect the population from water-borne diseases



- **DRAINAGE:** Manage the large volumes of water from rainfall by collecting it and conveying it out of the city efficiently to avoid flooding, damage to properties

SUPPLY SCALE

▪ Macro-scale:

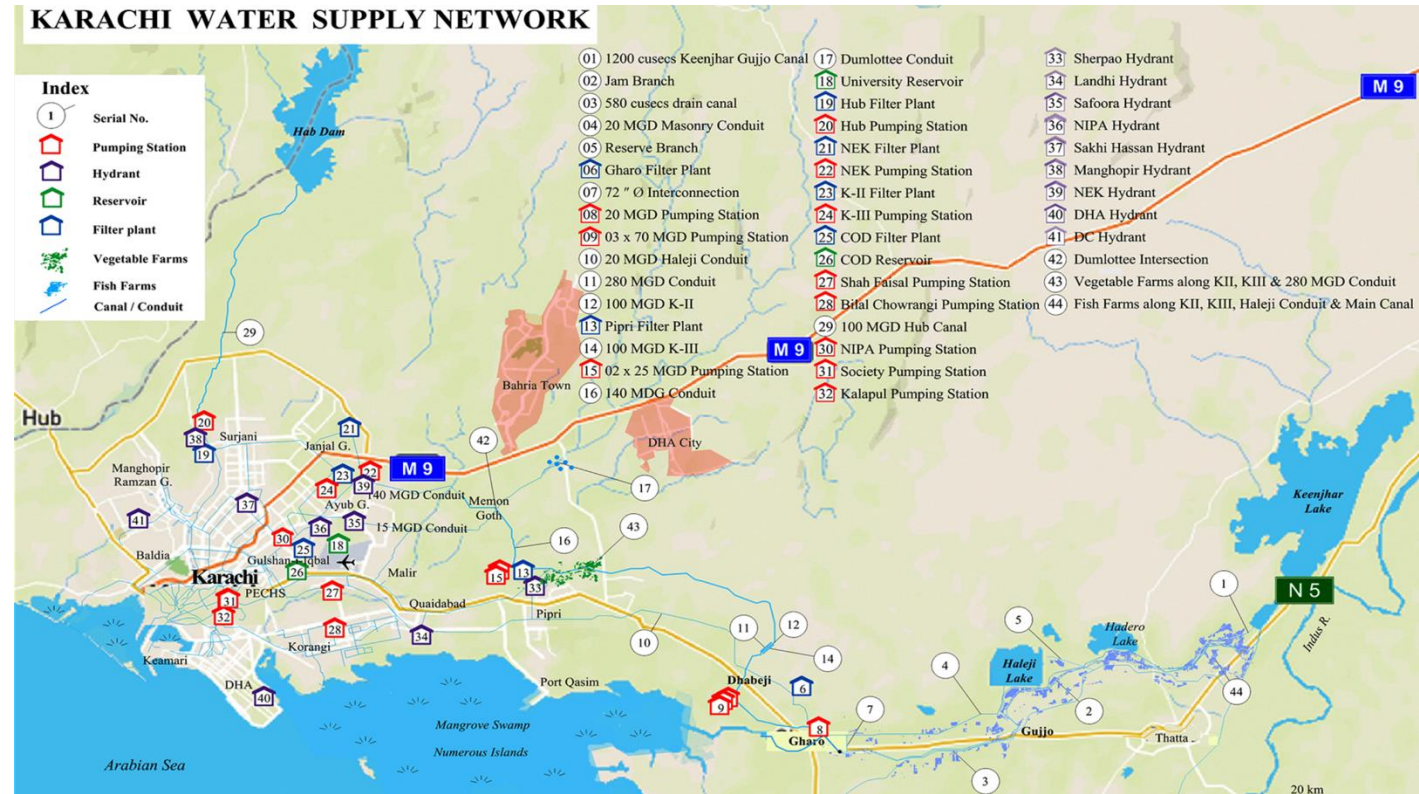
- Reservoirs, large mains, inter-annual transfer of storage
- Bulk water supply sourcing
- Environmental flows management
- Billing of water utilities

▪ Meso-scale:

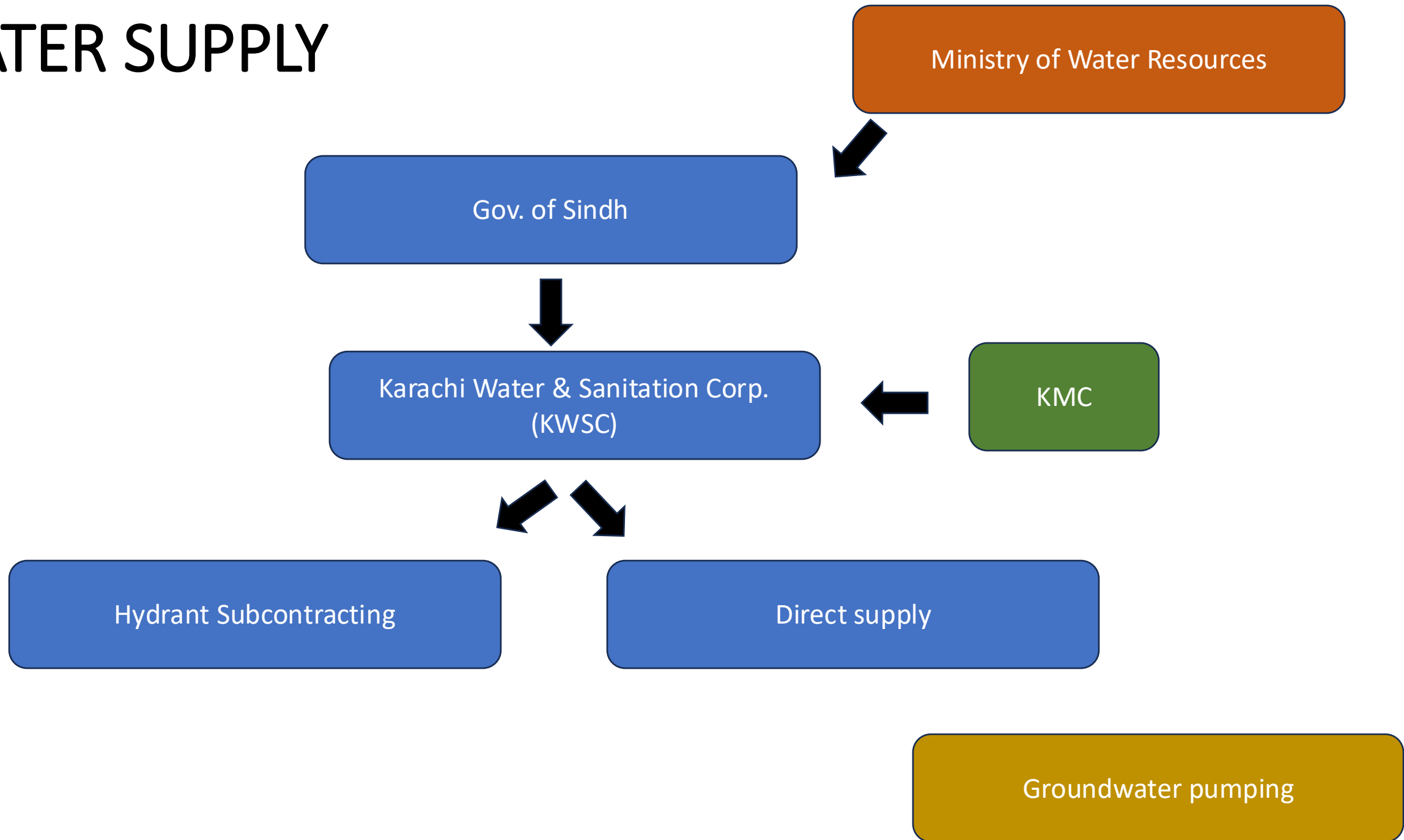
- Disinfection & distribution through local reticulation network
- Operation & maintenance of local assets
- Local-level water recycling
- Billing of consumers

▪ Micro-scale:

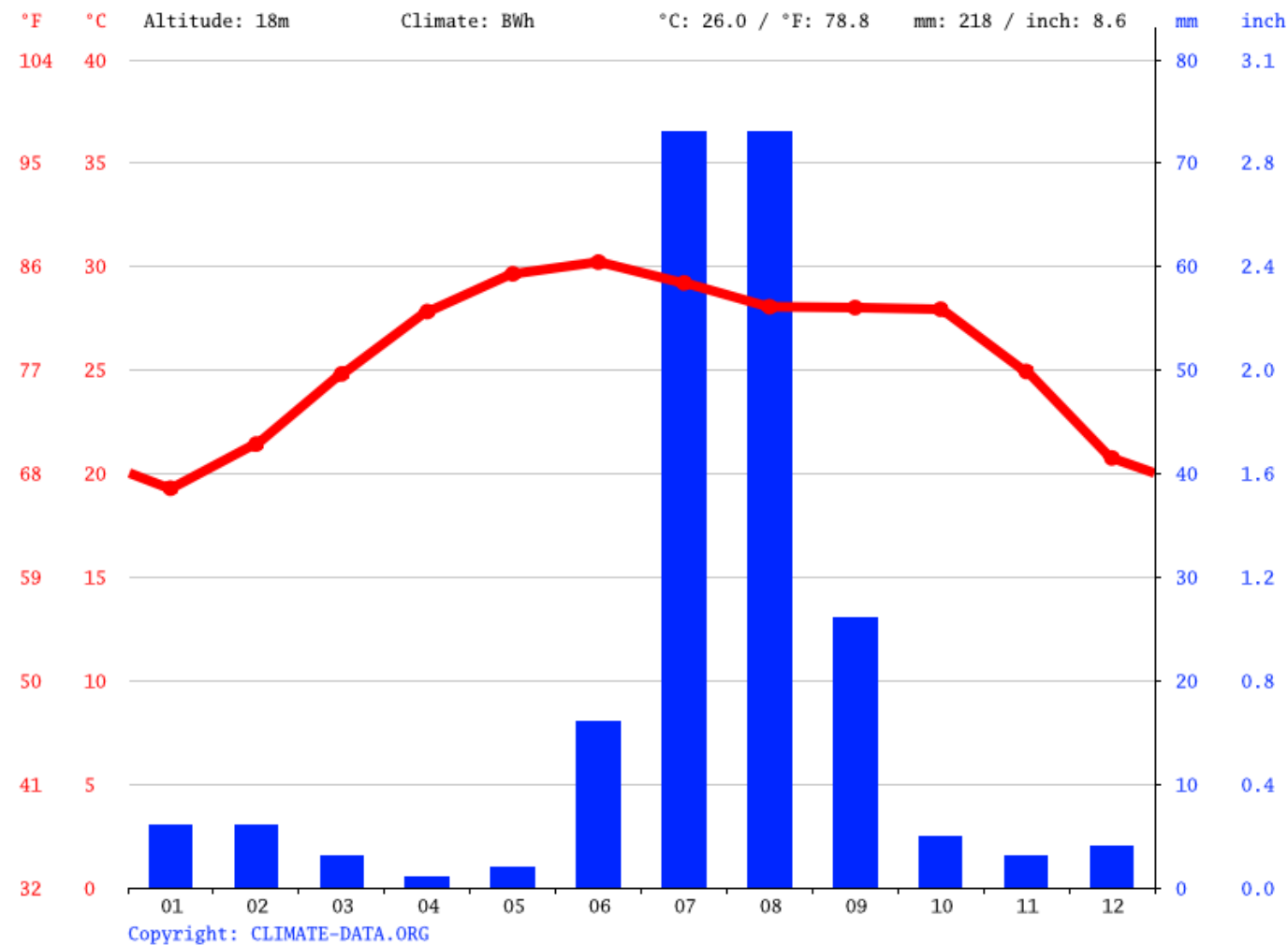
- Consumption management
- Harvesting of rainwater/greywater/stormwater



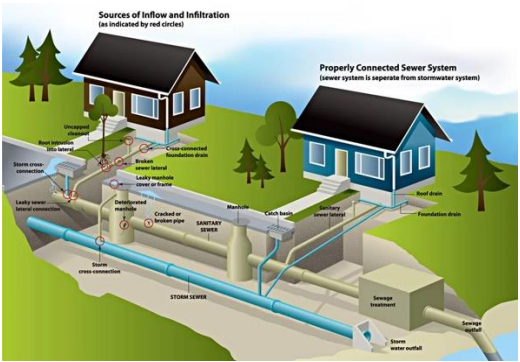
WATER SUPPLY



Karachi Rain patterns



SANITATION

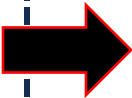


WASTEWATER GENERATION

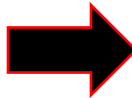
House

Industry

Agri

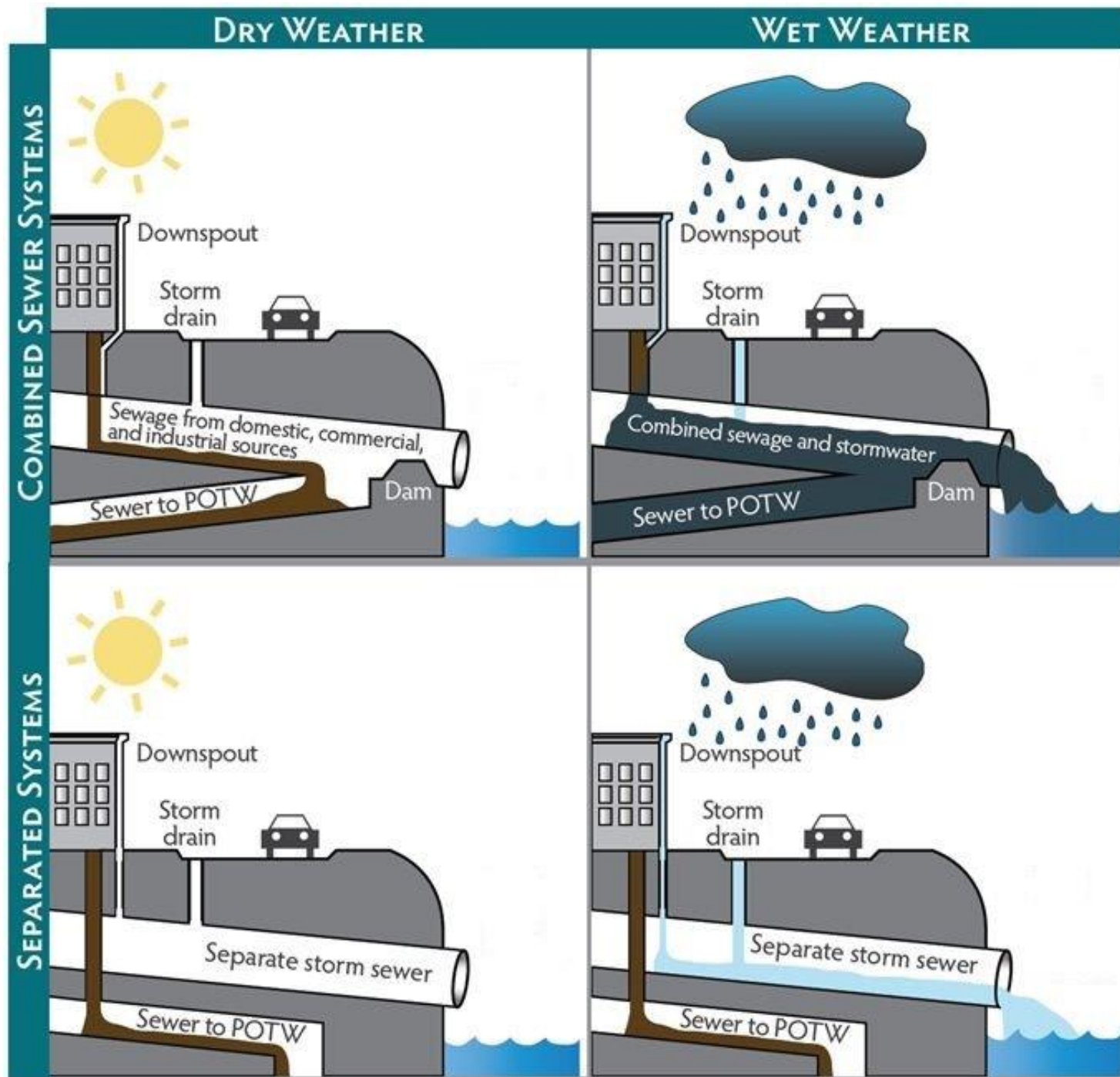


WASTEWATER TREATMENT



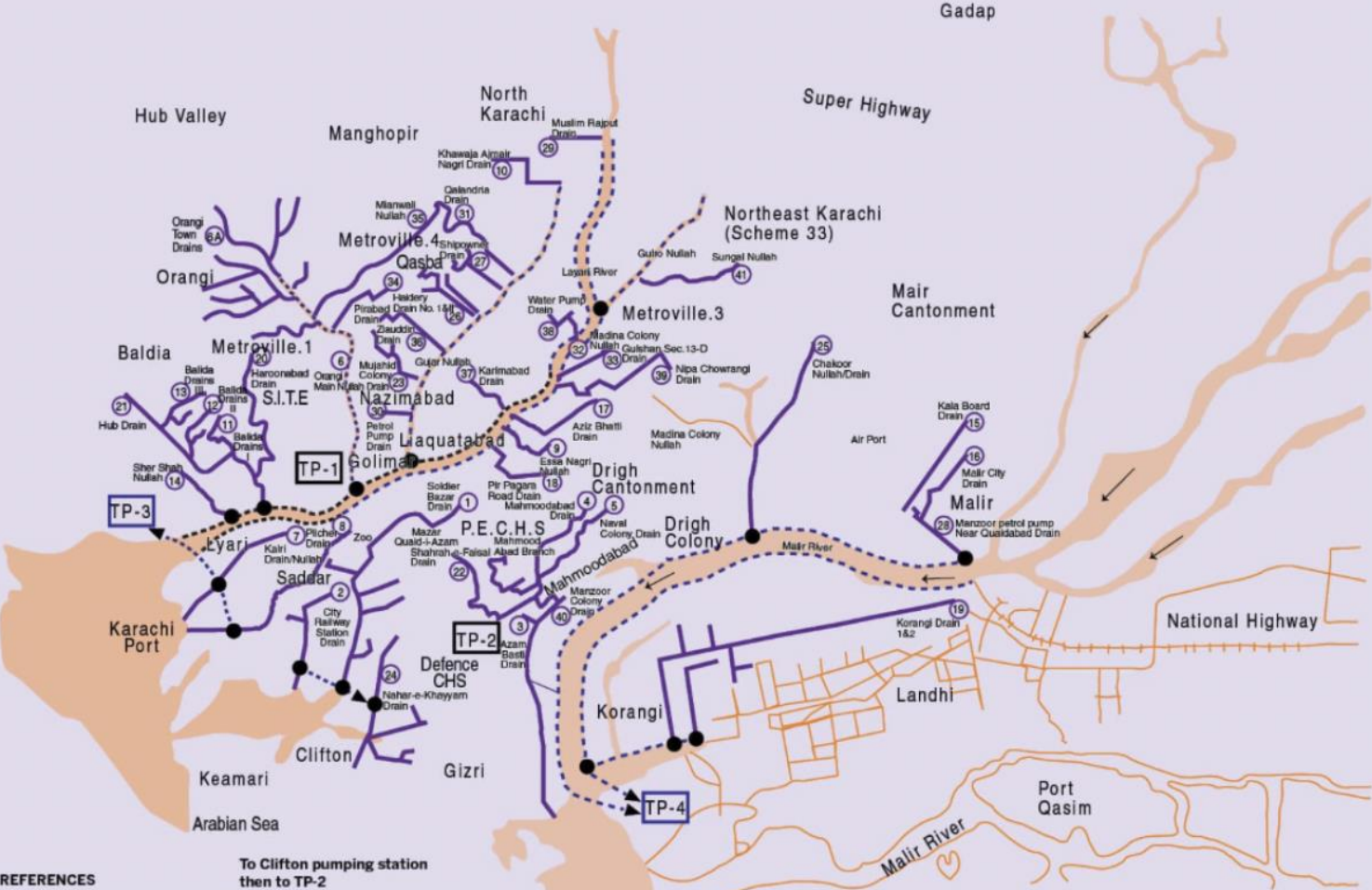
WATER
FOR REUSE





Karachi map showing the documented natural *nullahs* and drains (which are the main disposals for sewage and rain water) and the intergration of these with the mains and STPs.

CONCEPTUAL PLAN FOR A REALISTIC SEWAGE DISPOSAL SYSTEM



REFERENCES

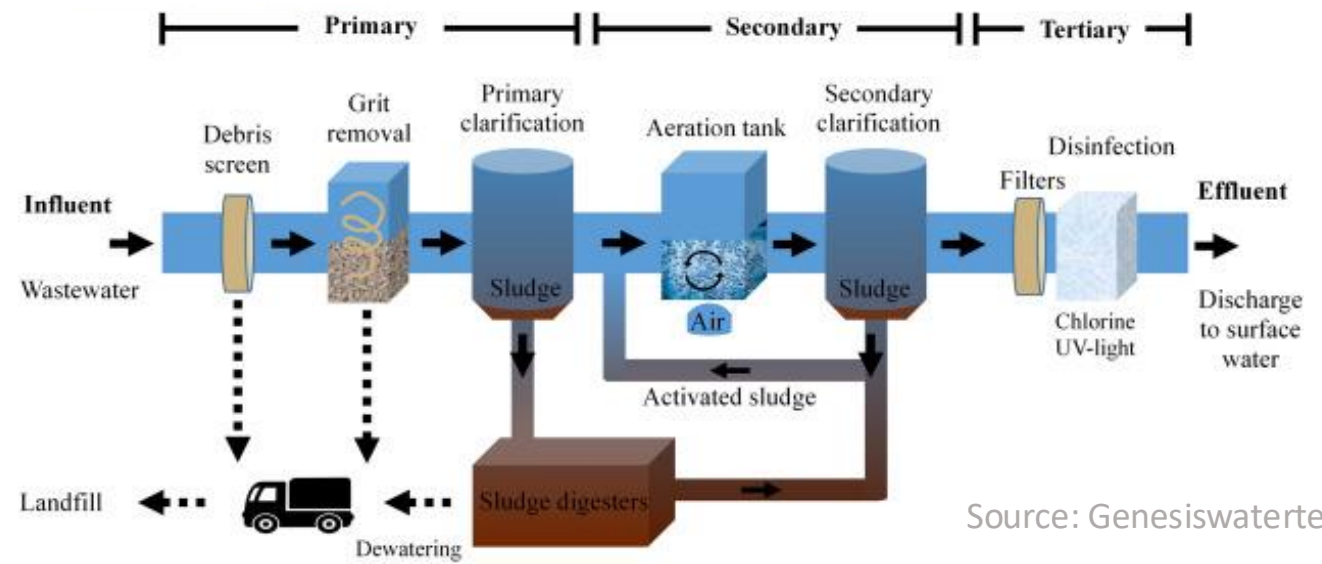
EXISTING	PROPOSED
Main natural Nullah	Interceptors/box trunk
Branch Nullah	Sewage treatment plant
Drain	Intake chambers
Drain/Nullah's Number	
Interceptors/box trunk	
Sewage treatment plant	

S.No.	Name of Drain/Nullah	Length (ft)
1.	Soldier Bazar Drain +4 Branches	40000
2.	City Railway Station Drain	10825
3.	Azam Basti Drain	866
4.	Mahmoodabad Drain & Branch	27000
5.	Naval Colony Drain	11258
6.	Orangi Nullah Main (Banaras Chowk to Lyari Nadi)	19918
6A.	Orangi Town Drain	94234
7.	Kalri Drain / Nullah	9526
8.	Pitcher Drain	14700
9.	Essa Nagri Nullah And Welfare Colony Drain	7794
10.	Khawaja Ajmair Nagri Drain And North Karachi Sec. 11-A Drain	14000
11.	Baldia Drains – I	15715
12.	Baldia Drains – II	7000
13.	Baldia Drains – III	7784
14.	Sher Shah Nullah	6603
15.	Kala Board Drain	9310
16.	Malir City Drain	9959
17.	Aziz Shatti Drain	13856
18.	Pir Pagara Road Drain	8877
19.	Korangi Drain 1 & 2	48496
20.	Haroonabad Drain	17753
21.	Hub Drain	23815
22.	Shahrah-e-Faisal Drain	6000
23.	Mujahid Colony Drain	12500
24.	Nahar-e-Khayyam Drain	5629
25.	Chakoor Nullah/Drain	23080
26.	Haidery Drain I & II	15600
27.	Shipowner College Drain	6575
28.	Manzoor Petrol Pump to Islamia Kanta Malir (Near Qaidabad) Drain	900
29.	Muslim Rajput Colony Drain	2552
30.	Petrol Pump Drain	3350
31.	Qalandria Drain	17200
32.	Madina Colony Nullah	1299
33.	Gulshan-e-Iqbal Sec. 13-D Drain	9093
34.	Pirabad Drain	2598
35.	Mianwali Colony Nullah (Joins 6A)	11238
36.	Ziauddin Hospital Drain	16478
37.	Karimabad Drain	7000
38.	Water Pump Drain	5890
39.	NIPA Chowrangli Drain	15588
40.	Manzoor Colony Drain	11700
41.	Sungel Nullah	8660

Source: OPP-RTI

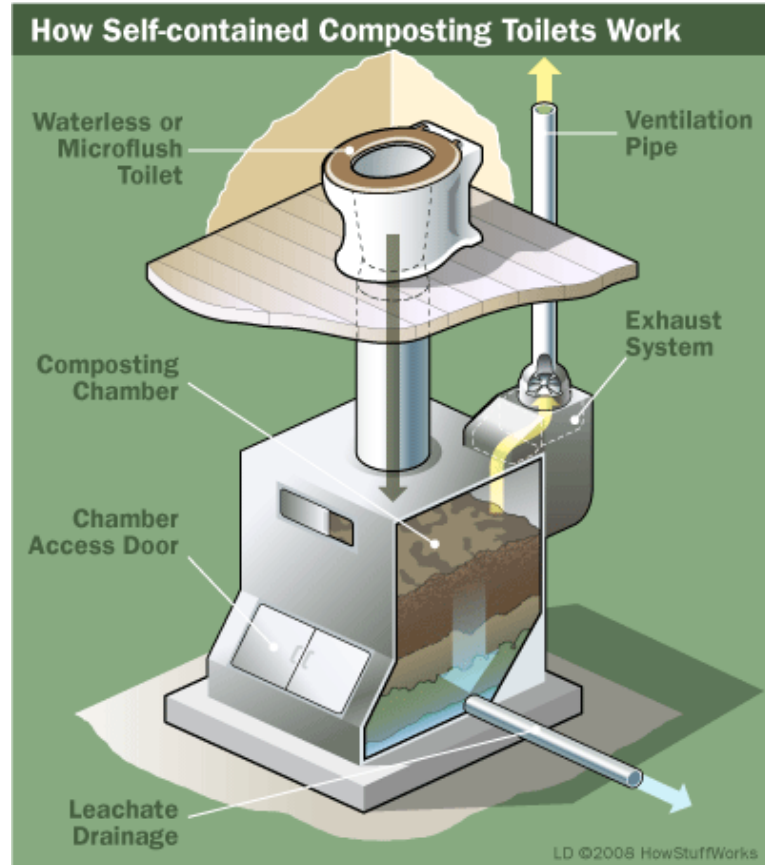
WASTEWATER TREATMENT (Macro scale)

- Treatment of Water in different stages (Primary/Secondary/Tertiary)\
 - **Primary:** Initial screening, removal of gross solids/sediment
 - **Secondary:** Biological treatment, removal of nutrients
 - **Tertiary:** More advanced treatment processes (targeting difficult chemicals/microbes)
- Various options for final effluent:
 - Discharge into receiving water bodies (strict environmental regulatory guidelines)
 - In-direct / Direct Reuse
- Sludge Management
 - Solids captured from Wastewater Management has to be dealt with
 - Further treatment process for solids and subsequent disposal or agricultural application

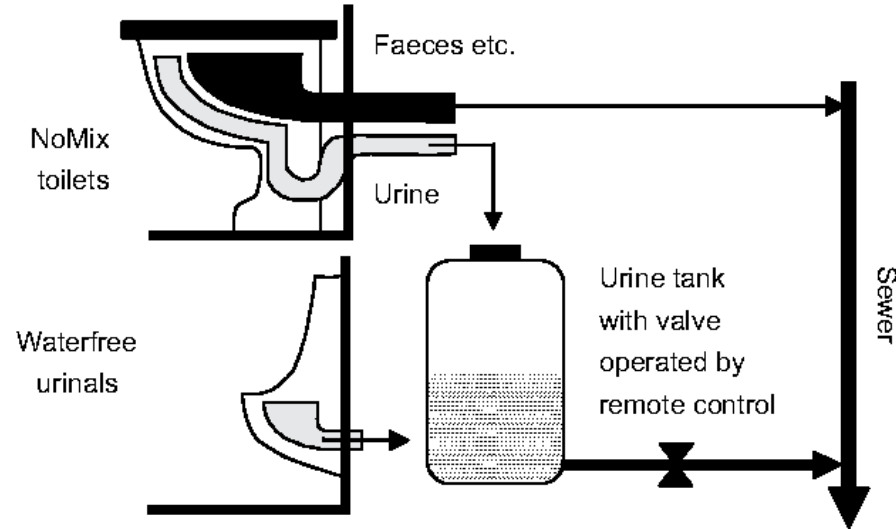


Source: Genesiswatertech

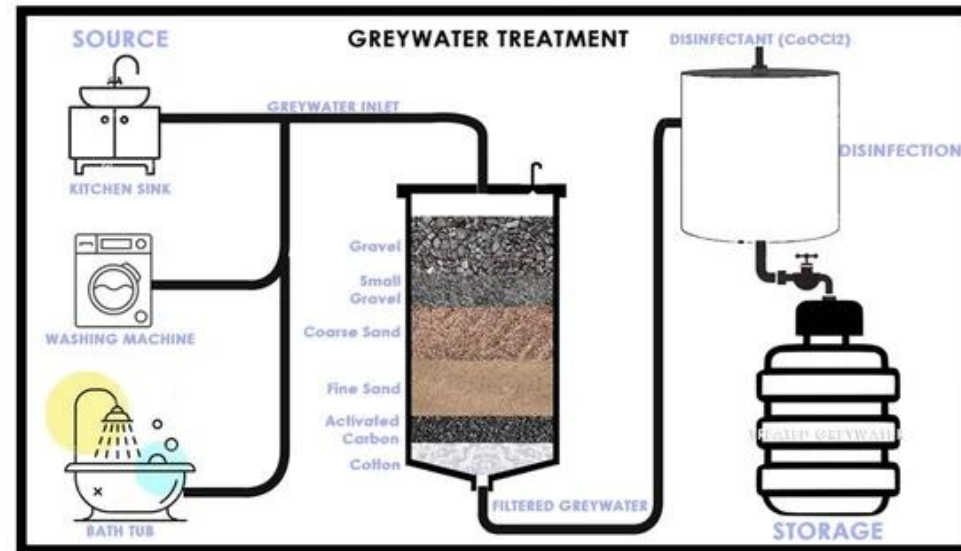
WASTEWATER TREATMENT (Micro scale)



1 COMPOSTING TOILETS



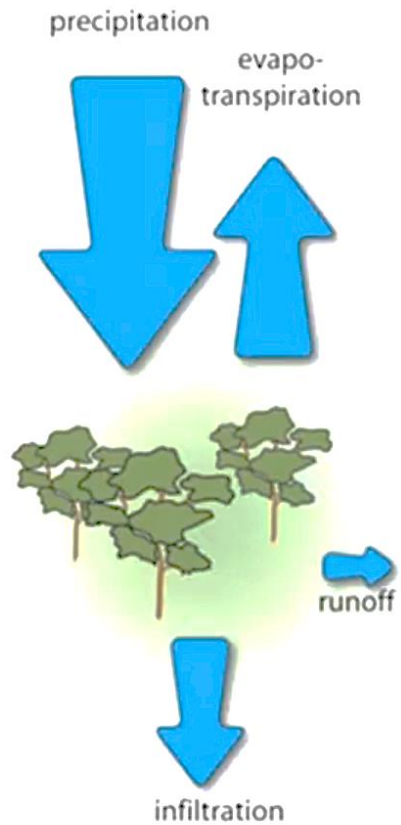
2 SOURCE SEPARATION



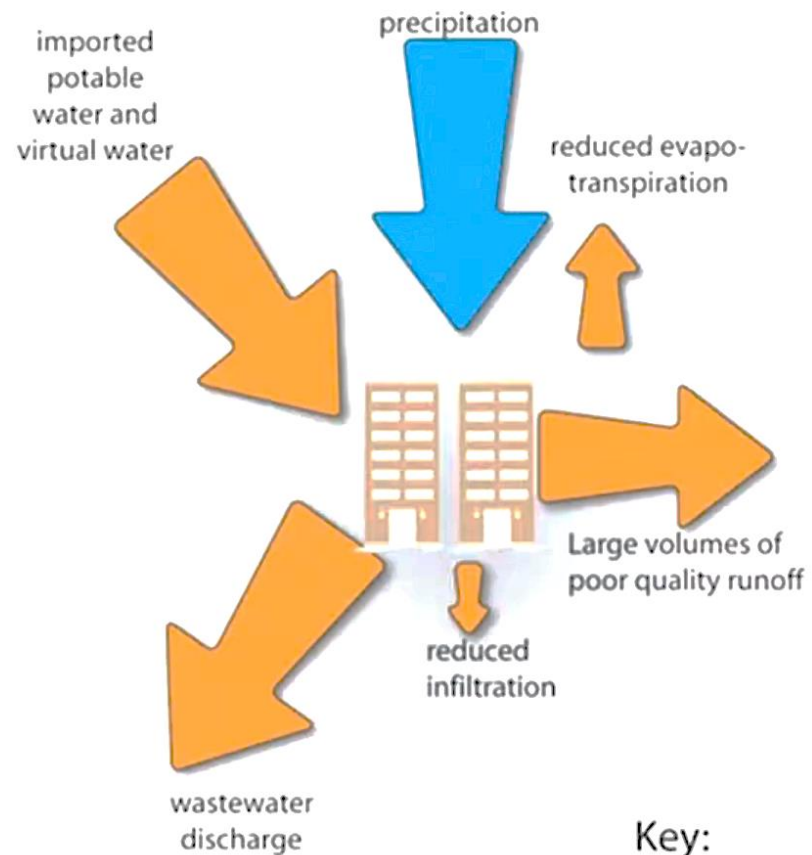
3 GREYWATER TREATMENT

DRAINAGE

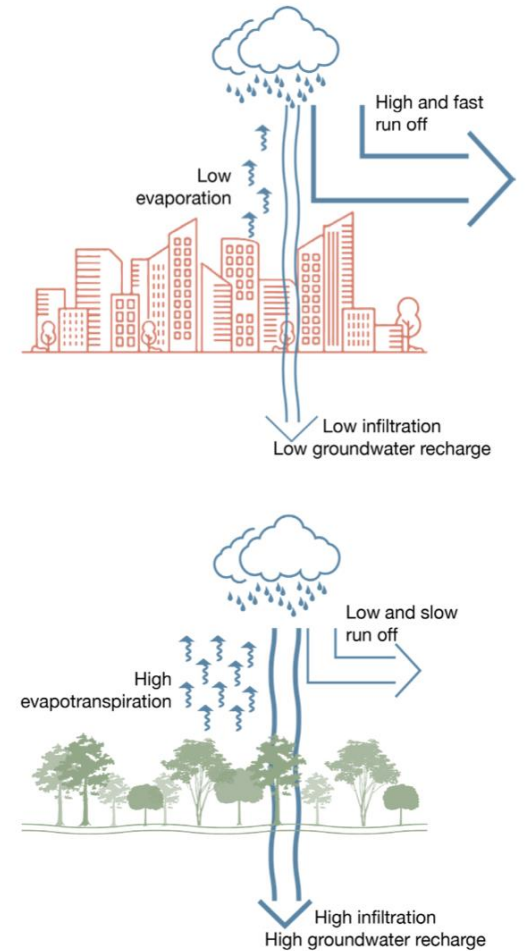
natural water balance



Urban water balance

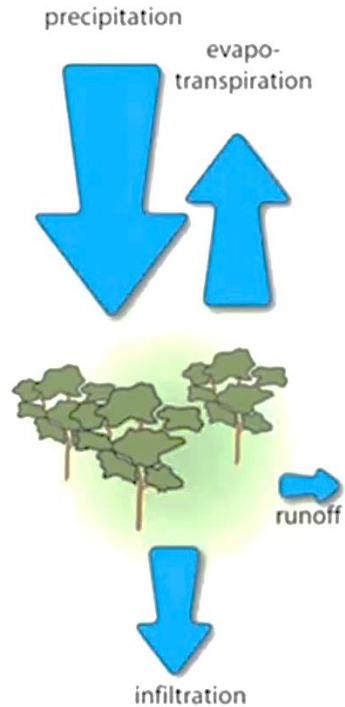


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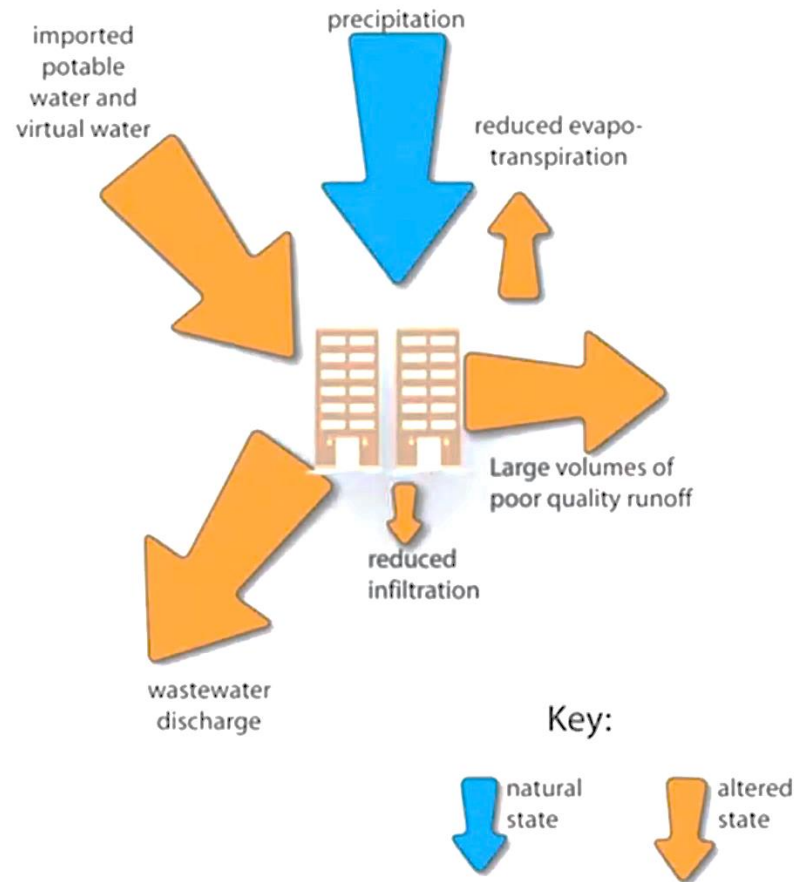


DRAINAGE

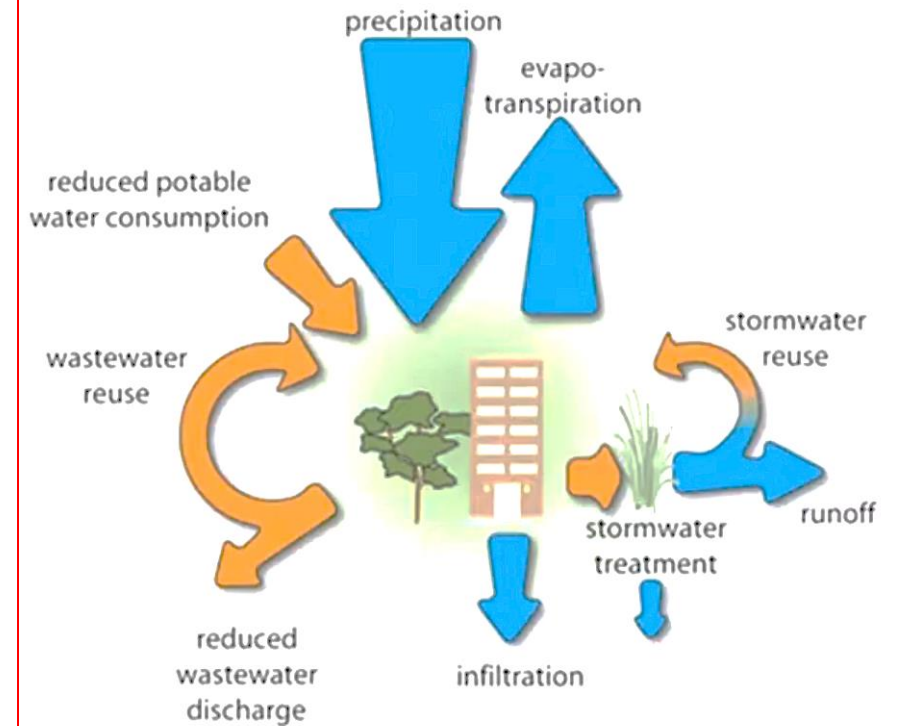
natural water balance



Urban water balance



WSUD water balance



WSUD SOLUTIONS



Rainwater harvesting



Green facade



Bio-retention areas



Permeable paving



Biotopes



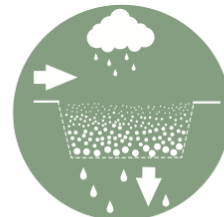
Infiltration zones and
trenches



Open stormwater
canals/ drains



Active
evapotranspiration



Gravel or sand filters



Swales



Passive
evapotranspiration



Rooftop retention



Geo-cellular system